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Control Chart Tests Based on Geometric Moving Averages

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A geometrical moving average gives the most recent observation the greatest weight, and all previous observations weights decreasing in geometric progression from the most recent back to the first. A graphical procedure for generating geometric moving averages is described in which the most recent observation is assigned a weight r. The properties of control chart tests based on geometric moving averages are compared to tests based on ordinary moving averages.

1. INTRODUCTION

An $\bar{X}$ control chart presents a graphical summary of information collected for the purpose of providing continuing information on the functioning of a process. This information is useful in indicating the existence, and in some instances the causes, of conditions requiring corrective action. A central line is drawn on the control chart to represent the level at which the process operates in the absence of trouble, and limit lines are placed at prescribed distances above and below this central line. The presence of trouble is indicated if the point $\bar{X}_j$, representing the average of several measurements taken at time j ($j = 1, 2, 3, \cdots$), falls outside either of the limit lines. (Symbols such as $\bar{X}_j$ are used to denote both a point and its scale value). In practice, corrective action is sometimes taken when a run of points occurs that is highly improbable in the absence of trouble, though no single point falls outside the limit lines; for example, a run of seven consecutive points on one side of the central line is sometimes taken as an indication that trouble is present.

In a recent paper concerned primarily with the properties of run tests in detecting shifts in a process average, (1) the writer indicated that tests based on moving averages should be more efficient than run tests in making use of the information supplied by control chart points. A graphical procedure was outlined for generating points representing moving averages of 2, 4, 8 or 2^i consecutive points. The procedure was based on the fact that the point half-way between two points on their connecting straight line represents their average value. For example, points representing the average of eight $\bar{X}$'s are obtained from points representing two sets of four $\bar{X}$'s, which in turn are obtained from points representing pairs of $\bar{X}$'s. The necessity of generating the auxiliary points representing four and two $\bar{X}$'s is a disadvantage; such points could be incorporated into the testing procedure by drawing limit lines for them but it is questionable whether the improvement in statistical properties would justify the added complexity.

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J. W. Tukey suggested to the writer an alternative graphical procedure that requires no auxiliary points for generating a type of moving average in which the entire history of $\bar{X}$'s is assigned weights, with weights decreasing as a geometric progression from the most recent point back to the first. Such moving averages are called geometric moving averages in this paper, while the term moving average without the adjective "geometric" refers to the ordinary moving average in which k consecutive points are assigned equal weights of $1/k$. In computing a geometric moving average the most recent point is assigned a weight of r, $0 < r \leq 1$, and every other point is assigned a fraction $(1 - r)$ of the weight of its immediate successor.

This paper outlines a procedure for generating geometric moving averages, and compares the properties of control chart tests based on such averages with tests based on ordinary moving averages.

2. Generating Geometric Moving Averages

Let Z_i denote the geometric moving average at time j ($j = 0, 1, 2, 3, \cdots$). It is convenient to set $Z_0 = \mu_0$, where μ_0 is the ordinate of the central line of control chart. Thus,

$$\begin{aligned} Z_0 &= \mu_0 , \\ Z_i &= (1 - r)Z_{i-1} + r\bar{X}_i , \qquad j > 0. \end{aligned} \tag{1}$$

The point Z_i is located a fraction r of the distance from Z_{i-1} to $\bar{X}_i$ on their connecting straight line.

To exhibit the geometric progression of weights assigned to the $\bar{X}$'s, (1) can be written as

$$\begin{aligned} Z_i = (1 - r)^j\mu_0 + r(1 - r)^{i-1}\bar{X}_1 + r(1 - r)^{i-2}\bar{X}_2 \\ + \cdots + r(1 - r)\bar{X}_{i-1} + r\bar{X}_i , \qquad j \geq 0. \end{aligned} \tag{2}$$

Rearranging terms, this becomes

$$\begin{aligned} (Z_i - \mu_0) = r(1 - r)^{i-1}(\bar{X}_1 - \mu_0) + r(1 - r)^{i-2}(\bar{X}_2 - \mu_0) \\ + \cdots + r(1 - r)(\bar{X}_{i-1} - \mu_0) + r(\bar{X}_i - \mu_0), \qquad j \geq 0, \end{aligned} \tag{3}$$

and clearly the expected value of Z_i is μ_0 when the expected values of the $\bar{X}$'s are all equal to μ_0. Then, if the $\bar{X}$'s are independent and have a common standard deviation $\sigma_{\bar{x}}$, the standard deviation of Z_i is

$$\sigma_{Z_i} = \sqrt{\frac{r}{2 - r}\left[1 - (1 - r)^{2i}\right]}\,\sigma_{\bar{x}} , \tag{4}$$

which increases to its limiting value

$$\sigma_Z = \sqrt{r/(2 - r)}\,\sigma_{\bar{x}} \tag{5}$$

as j increases.

To generate the Z's, proceed as follows:

1. Mark point $\bar{X}_1$ at abscissa 1.
2. Mark $Z_0 = \mu_0$ at abscissa $1 - 1/r$.

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3. Draw a straight line connecting Z_0 with $\bar{X}_1$; the point on this line with abscissa $1 - (1 - r)/r$ is Z_1 .

The next point, Z_2 , will be at abscissa $2 - (1 - r)/r$ on the straight line connecting Z_1 with $\bar{X}_2$, which is plotted at abscissa 2. Both the $\bar{X}$'s and the Z's progress one abscissa unit at a time; Z_i is always $(1 - r)/r$ abscissa units to the left of $\bar{X}_i$.

Figure 1 illustrates the procedure for the case where $r = \frac{2}{5}$, which is a value of r that may have wide appeal in practice. For one thing, with $r = \frac{2}{5}$ the Z's fall on vertical lines drawn half-way between vertical lines of the $\bar{X}$'s, thus minimizing confusion between the two types of points. Also, with $r = \frac{2}{5}$, $\sigma_Z = \sigma_{\bar{X}}/2$ so that, as shown in Figure 1, 3σ limit lines for the Z's are half as far from the central line as 3σ limit lines for the $\bar{X}$'s.

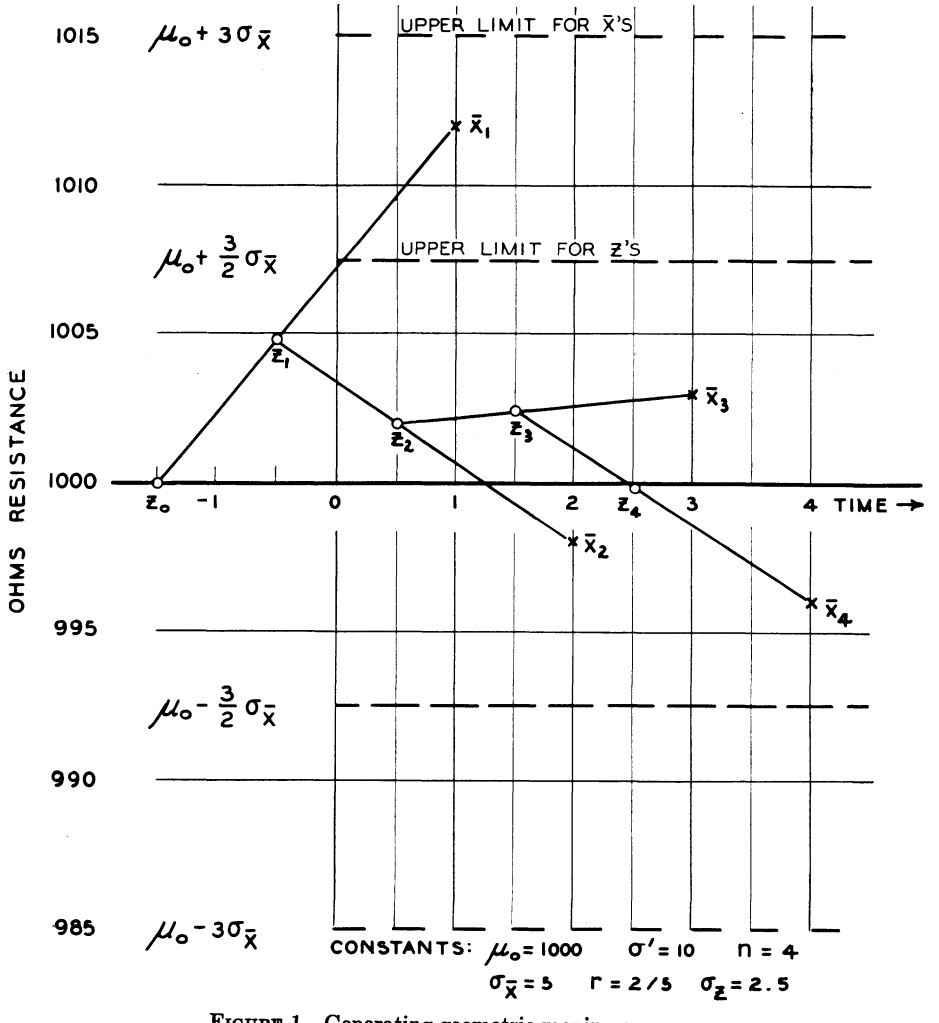


FIGURE 1—Generating geometric moving averages.

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Limit lines for the Z's should be based on σ_Z ; i.e., place limit lines at $\mu_0 \pm 3\sigma_Z$. In general it wouldn't pay to widen the limits with increasing j using equation (4).

The sequence of Z's may be started at any time j simply be setting $Z_{j-1} = \mu_0$ and following the procedure outlined above. An error ϵ in Z_{j-1} manifests itself as an error $\epsilon(1 - r)^k$ in Z_{j-1+k} .

3. Process Model

One way of comparing control chart procedures is to assume that the end result is a sequence of decisions on whether or not to take action. An evaluation of the effectiveness of various tests must center attention on the relative frequencies of two types of decision errors, with their attendant economic consequences. In discussing the two types of decision errors it is convenient to adopt the terminology of hypothesis testing and define a null hypothesis; then erroneous rejections and acceptances of the null hypothesis are called, respectively, Type 1 and Type 2 errors.

It is assumed that at any time there is a process average $\mu = \mu_0 + \delta$, where μ_0 is the nominal process average and δ, which is measured in the same units as μ_0 , is the difference between the current process average and its nominal value. In the absence of trouble $\delta = 0$.

The testing procedure consists of periodically testing the null hypothesis that $\mu = \mu_0$, or, equivalently, that $\delta = 0$. A rejection of the null hypothesis when it is "true" is a Type 1 error; such an error results in wasted effort of looking for nonexistent trouble. An acceptance of the null hypothesis when it is not true is a Type 2 error; the cost of such an error must be expressed in terms of the relatively poor quality of output from time j to time $j + 1$, when the null hypothesis is next tested.

The over-all average quality of the product depends on the way the process average varies with time, as affected by occurrences of trouble and by attempts to keep the process free of trouble. A simplifying concept is that the process average remains at its nominal level until trouble shifts it to a new level where it remains until the trouble is eliminated.

Let the random variable y be the number of points following the occurrence of trouble until the decision is made that trouble is present. A sequence of $y - 1$ consecutive Type 2 errors is made. The distribution function of y depends not only on the amount δ of the shift but also on the standard deviation of $\bar{X}$. For this reason $\Delta = \delta/\sigma_{\bar{X}}$ is used, rather than δ, as a measure of the amount of the shift.

For most practical purposes the distribution function of y is adequately summarized by its expected, or average, value $E(y)$. Curves of $E(y)$ versus Δ are presented in Figures 2–5 to describe the effectiveness of particular tests in detecting shifts in a process average. The test notation is defined in Section 4.

Since $\bar{X}_j$ is the average of n measurements taken at time j, the assumption is made that its distribution is normal with average $\mu = \mu_0 + \delta$ and standard deviation $\sigma_{\bar{X}} = \sigma'/\sqrt{n}$, where σ' is the standard deviation of the individual measurements and is assumed to be known from past experience and not subject to change. In considering a curve of $E(y)$ versus Δ for a particular application,

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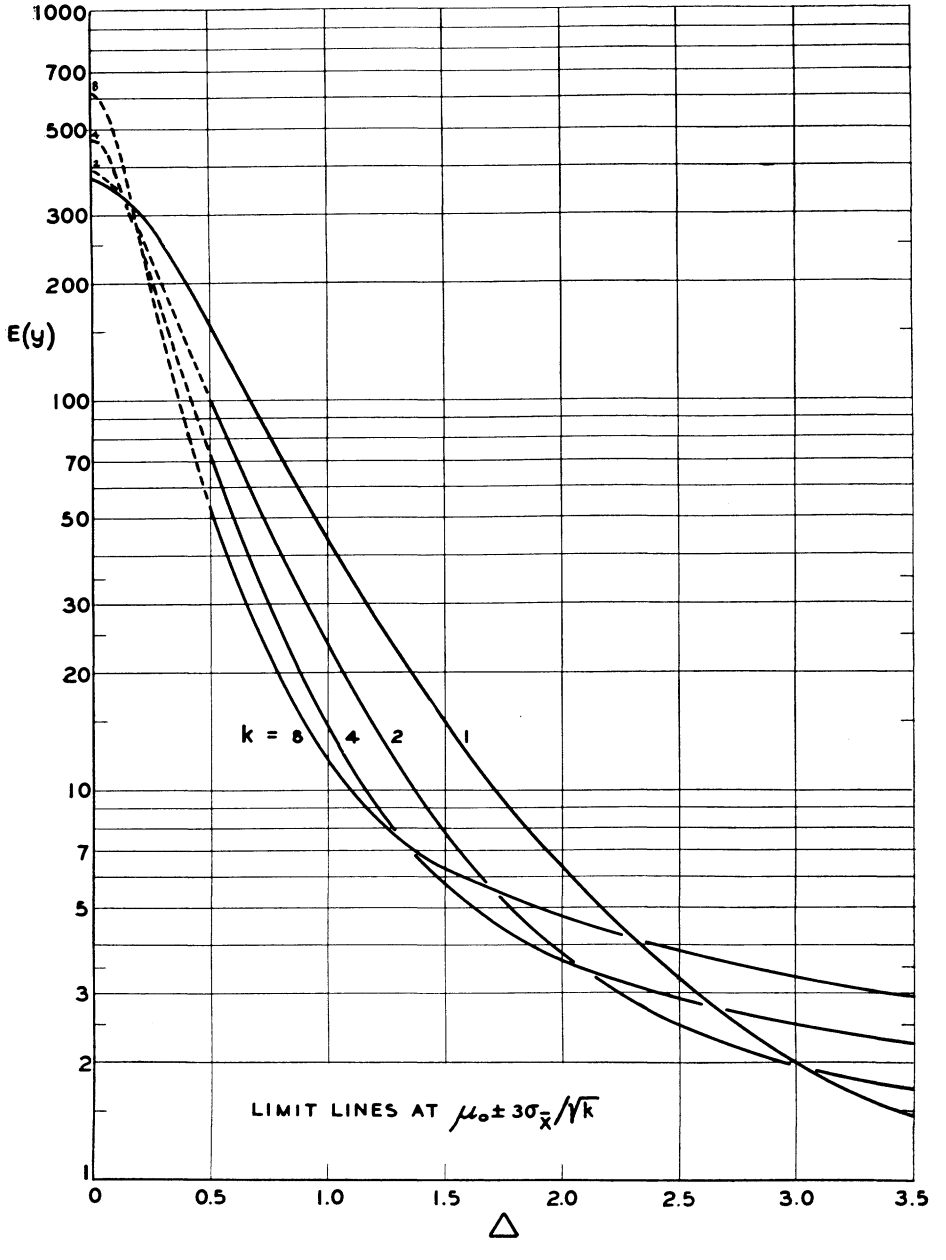


FIGURE 2—Moving average tests, M_k.

note that $\Delta = \delta\sqrt{n}/\sigma'$; thus, while the curves are independent of sample size n, the value of Δ applicable in a particular situation depends on n.

For a particular value of Δ, the corresponding value of $E(y) - 1$ is the expected number of consecutive Type 2 errors made following a shift of the magnitude associated with Δ. The value of $E(y)$ at $\Delta = 0$ is a measure of the frequency of

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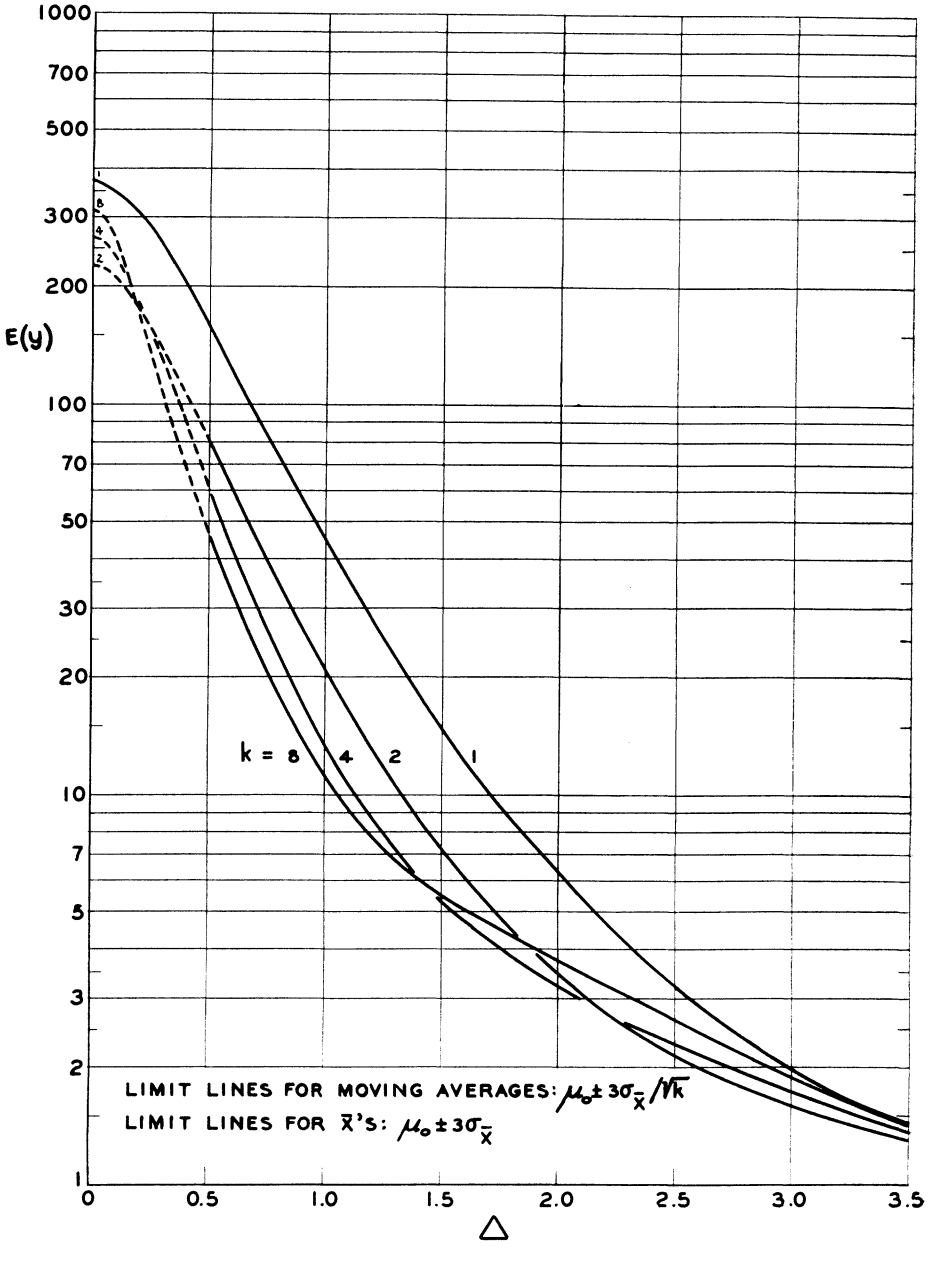


FIGURE 3—Moving average tests, $M_{1,k}$.

Type 1 errors. For example, the top (for central values of Δ) curves in Figures 2–5 pertain to the standard control chart test with 3σ limits, and on these top curves the value of $E(y)$ at $\Delta = 0$ is roughly 370; this means that when the process average is at its nominal level, μ_0 , an average of 370 points can be expected between successive Type 1 errors when using the standard control chart test.

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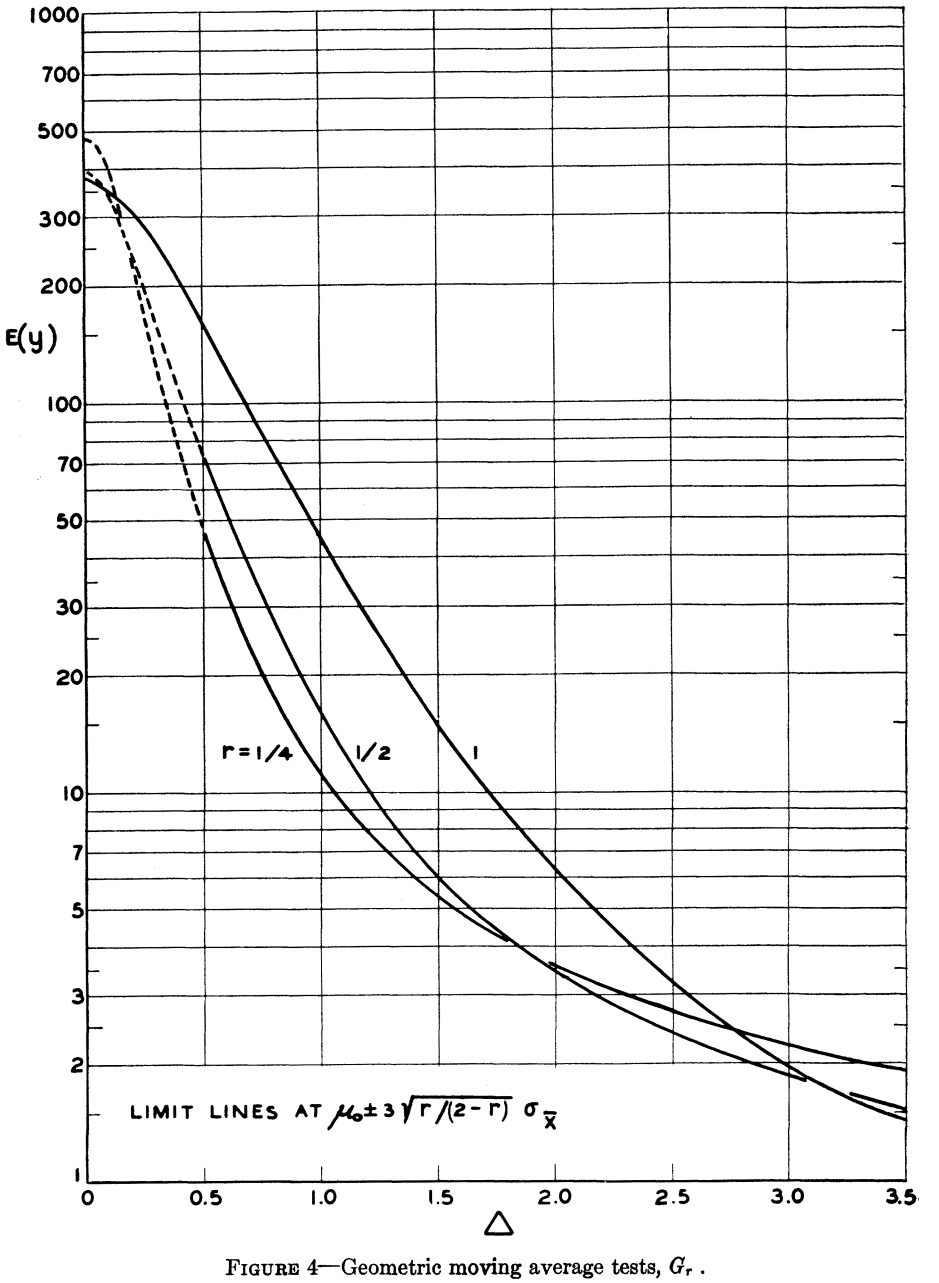


FIGURE 4—Geometric moving average tests, G_r .

4. CURVES OF $E(y)$ VERSUS Δ

Let M_k denote the test that prescribes rejection of the null hypothesis at time j if the moving average of k consecutive $\bar{X}$'s falls outside limit lines at $\mu_0 \pm 3\sigma_{\bar{X}}/\sqrt{k}$. (If limits other than 3σ limits are used for a statistic, the notation

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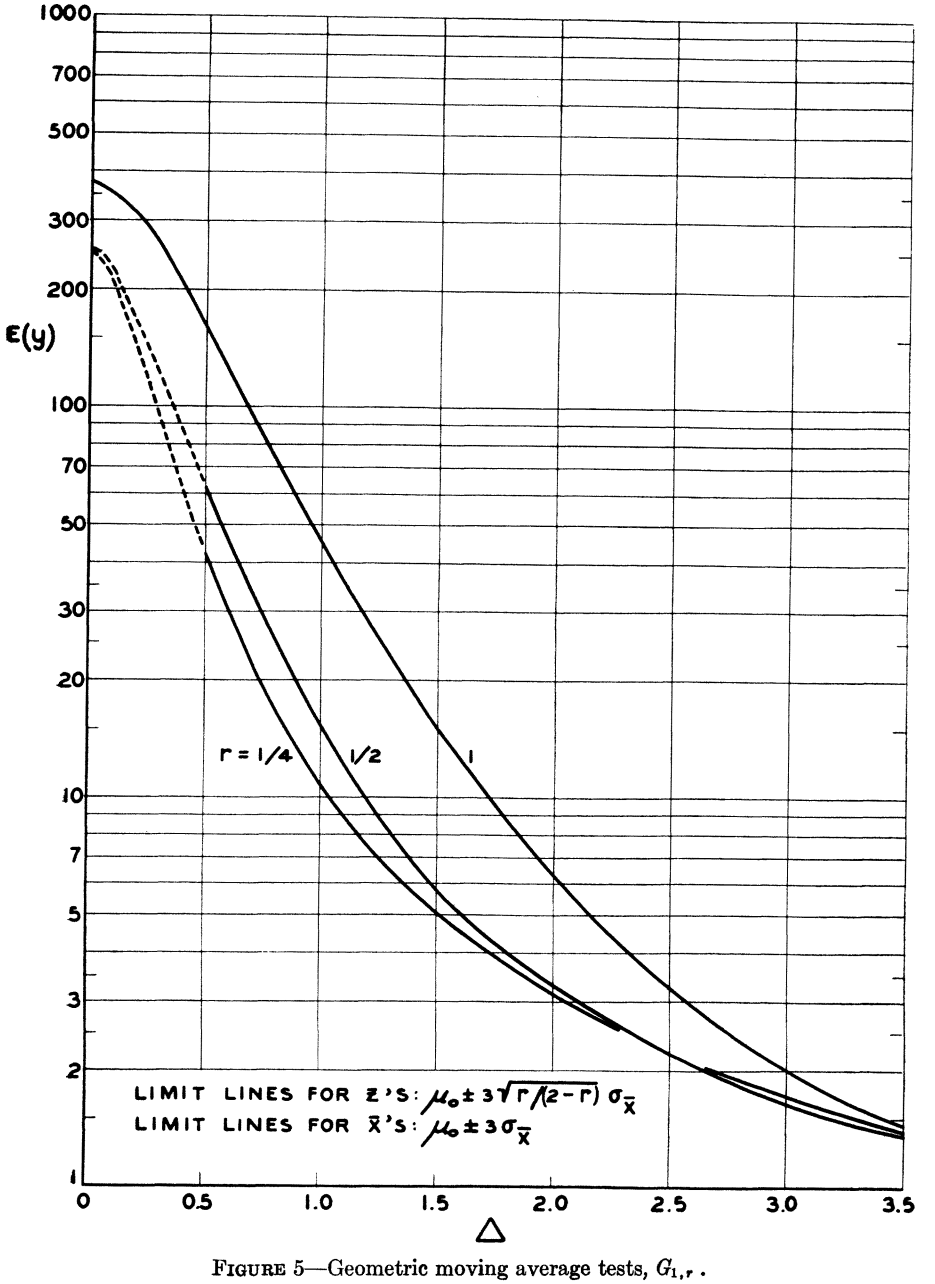


FIGURE 5—Geometric moving average tests, $G_{1,r}$.

will be supplemented to indicate the factor used—e.g., tests M_k (2) employ 2σ limits: $\mu_0 \pm 2\sigma_{\bar{x}}\sqrt{k}$.) Let $M_{1,k}$ denote a combination of this test with the standard control chart test, which prescribes rejection of the null hypothesis if $\bar{X}_i$ falls outside limit lines at $\mu_0 \pm 3\sigma_{\bar{x}}$. Similarly, let G_r denote the test prescribing rejection of the null hypothesis if the geometric moving average with weight

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parameter r falls outside limit lines at $\mu_0 \pm 3\sqrt{r/(2-r)}\ \sigma_{\bar{x}}$, and let $G_{1,r}$ denote such a test in combination with the standard control chart test. Figures 2 through 5 present curves of $E(y)$ versus Δ for these tests for various values of the weight parameters. Note in each case that when the weight parameters r or k are given their limiting value, 1, the test becomes the standard control chart test; comparisons with the standard test are thereby facilitated.

The curves are based largely on estimates of $E(y)$ at $\Delta = 0(.5)\ 3.5$ based on results of process simulations using 25,000 random normal numbers. Sets of eight numbers were used to represent the process prior to the occurrence of trouble; if the statistics computed with these eight numbers fell within their limits, an amount Δ was added to subsequent numbers and a count was made of the number y of such numbers until a statistic fell outside its limit. The average value of y for all such cycles (roughly $25{,}000/(E(y) + 8)$ in number) serves as an estimate of $E(y)$ for the particular test and Δ considered. Such estimates, other than at $\Delta = 0$, fall on or very close to the curves shown. The estimates at $\Delta = 0$ were adjusted to be consistent among themselves and with computed lower bounds.

While the curves of Figures 2–5 afford rough comparisons of the statistical properties of various tests under the conditions assumed, more useful comparisons could be made if the limit lines were set so that the tests had a common level of Type 1 errors. Such a comparison applied to the moving average tests of Figure 2 would show that each such test has its curve lower than the others in a particular interval of Δ, and these intervals approach $\Delta = 0$ for increasing k; consequently, it would be well to neglect the history prior to the k most recent control chart points in applications where interest is in Δ greater than a particular value Δ_k, which depends on k.

With geometric moving average tests, the entire history of points is given some weight; by considering only the k most recent points some improvement could be effected in detecting changes greater than a critical value determined by k. However, the more remote history is given relatively little weight, and has correspondingly little influence on test properties. In fact, it appears that given any moving average test a geometric moving average test can be selected that has roughly equivalent properties; the requirement for rough equivalence appears to be that the σ's be equal, i.e., that $r = 2/(k+1)$. The choice between two tests having similar statistical properties must be based largely on their relative operational simplicity in the application under consideration.

To illustrate the use of the curves, consider a resistor manufacturing process in which at the end of each hour four resistors are measured and the average of the measurements is plotted on an $\bar{X}$ control chart. Suppose experience indicates that $\mu_0 = 1000$ ohms and $\sigma' = 10$ ohms.

For test $G_{1,2/5}$ the limit lines for the $\bar{X}$'s are at $1000 \pm 3 \times 10/\sqrt{4}$ ohms, or at 985 and 1015, and the limit lines for the Z's are at 992.5 and 1007.5 (See Figure 1). Now suppose something happens to shift the process average to 1005 ohms. How long will it be, on the average, before this shift is detected?

Compute $\Delta = (1005 - 1000)/5 = 1$, and look at Figure 5. There is no curve for $r = \frac{2}{5}$, but an interpolation between values for $r = \frac{1}{4}$ and $r = \frac{1}{2}$ indicates that

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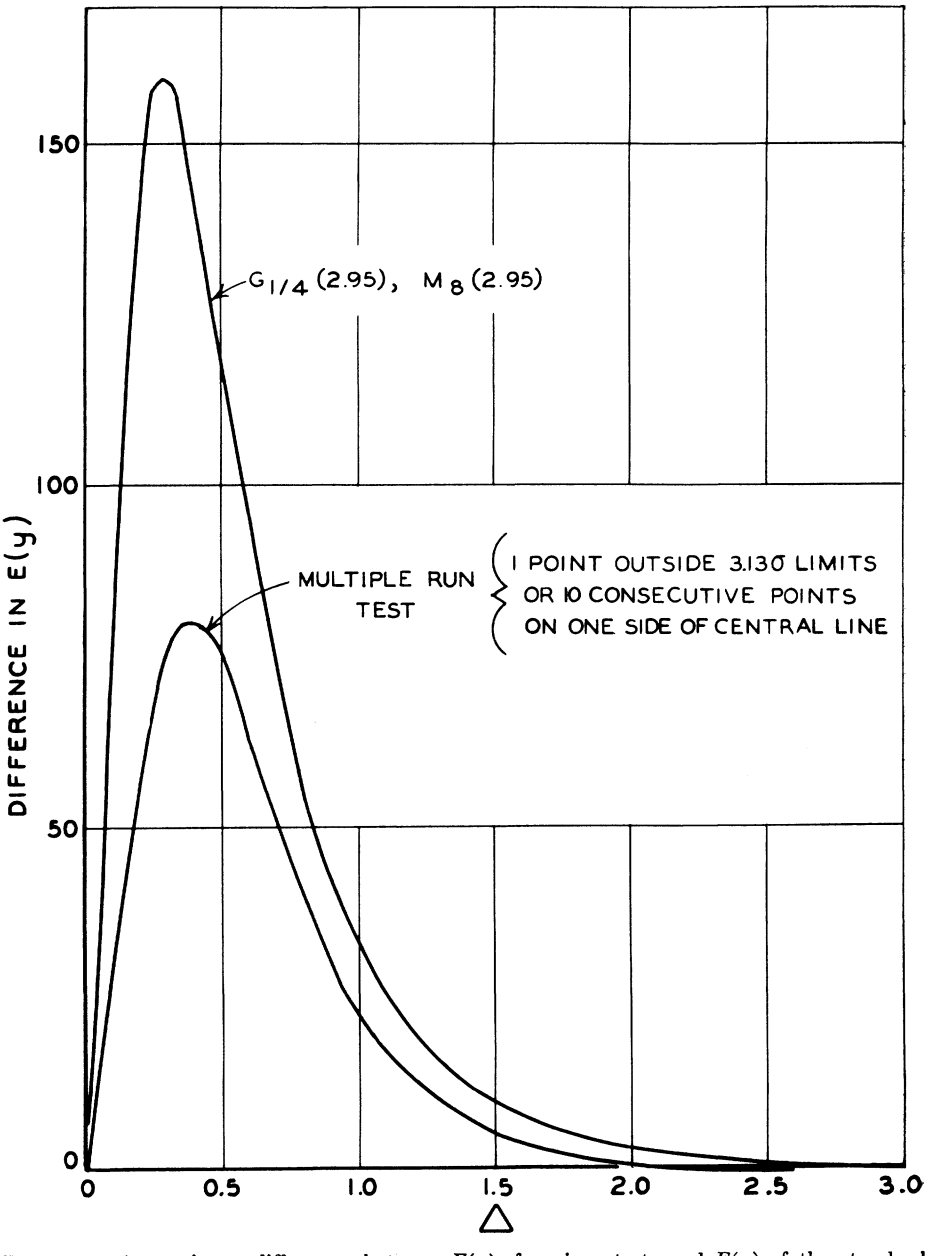


FIGURE 6—Approximate differences between $E(y)$ of various tests and $E(y)$ of the standard control chart test; arithmetic scale.

$E(y)$ is about 14. Thus, an average of about 14 pairs of $\bar{X}$'s and Z's will be plotted before either (1) an $\bar{X}$ is greater than 1015 (or less than 985) or (2) Z is greater than 1007.5 (or less than 992.5). The top curve of Figure 5 indicates that an average of about 44$\bar{X}$'s will be plotted before one of them is greater than 1015

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(or less than 985) when $\Delta = 1$. Since the top curve pertains to the standard control chart test, it appears that test $G_{1,2/5}$ will save an average of about $44 - 14 = 30$ hours in detecting a shift of five ohms in the process average in the example used here.

On the other hand, Figure 5 shows that $E(y)$ is about 250 at $\Delta = 0$ for test $G_{1,2/5}$, as compared with about 370 for the standard control chart test (top curve). Using test $G_{1,2/5}$, then, when the process remains trouble-free an average of 250 hours will pass between false indications of trouble (Type 1 errors) as compared with 370 hours for the standard control chart test.

Because the ordinate scales are logarithmic in Figures 2–5 it is difficult to appreciate the savings in $E(y)$ for the various tests as compared to the standard control chart test. The top curve in Figure 6 shows roughly the savings afforded using either $M_8(2.95)$ or $G_{1/4}(2.95)$, with limit lines set so that these tests have approximately the same frequency of Type 1 errors as the standard control chart test. Clearly, the important savings occur for relatively small shifts, say $|\Delta| < 2$, and the loss at large values of Δ is relatively small.

The curve of Figure 6 called "Run Test" is for a test that prescribes rejection of the null hypothesis if (1) a single $\bar{X}$ falls outside limit lines at $\mu_0 \pm 3.13\sigma_{\bar{X}}$ or (2) ten consecutive $\bar{X}$'s fall on one side of the central line of the control chart. The curve represents roughly what can be achieved using multiple run tests—use of additional run criteria would raise the curve in some ranges of Δ and lower it in others, provided the limit lines were set to yield the same frequency of Type 1 errors as the standard control chart test so that the curve would go through the origin of Figure 6.

5. Conclusion

Tests based on geometric moving averages compare most favorably with multiple run tests and moving average tests with regard to simplicity and statistical properties. The standard control chart test is simpler than alternative tests and cannot be improved on in detecting relatively large changes. In fact the parameters that would be chosen for the other tests in such applications would reduce such tests to the standard control chart test. In general, as interest centers on early detection of smaller and smaller changes, the appropriate parameter r of geometric moving average tests decreases from unity to smaller and smaller values, and the number k of consecutive points considered in moving average tests and multiple run tests increases.

The complexity of geometric moving average tests relative to the standard control chart test makes its use difficult to justify in many cases. In other cases the added sensitivity to relatively small changes in the process is adequate justification. In still other cases the cost of the added complexity is more than covered by a reduction in sampling costs. For example, in some applications a geometric moving average test used on a chart where the averages or medians of three measurements are plotted will be more effective than the standard control chart test based on the averages of four measurements.

Sample size and frequency of sampling have been largely eliminated from the discussion, since interest has been primarily on making efficient use of the informa-

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tion supplied by the control chart points. The preceding paragraph provides an example of the importance of considering these variables when selecting tests for particular applications.

Geometric moving average tests would seem to be particularly well suited to applications where tests based on ordinary moving averages are now used, including cases where the sample size is unity.

While this paper has been concerned specifically with $\bar{X}$ control charts, much of it could be rewritten word for word to apply to control charts of other sample statistics by simply replacing the symbol "$\bar{X}$" and the corresponding term "average" with appropriate substitutes. Certainly the particular weight patterns applied to the history of points on an $\bar{X}$ control chart in obtaining ordinary moving averages and geometric moving averages can just as readily and meaningfully be applied to the points on control charts of such useful sample statistics as the maximum, median, minimum and range. The curves of $E(y)$ versus Δ are generally applicable if allowances are made for the non-normality of the distributions of the sample statistics—curves pertaining to tests which apply weights to several control chart points are affected very little by non-normality.

It thus appears that the graphical procedure for applying geometric weight patterns may prove useful in a wide variety of applications.

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